(*symplecticness*) of the phase flow. We will later show (in Chap. 3) that such symplectic methods preserve a perturbed energy-like invariant (to very high accuracy); the preservation of this modified energy then ensures that the original energy is nearly conserved.

The methods considered in this chapter are for the most part very simple in terms of their composition and are based on principles of classical mechanics. The reader may be aware of methods for using variable stepsize, extrapolation, etc. For reasons discussed in the next chapter, these methods and the concerns that motivated their development, which are undoubtedly essential in many applications of ordinary differential equations, are less relevant in the setting of molecular dynamics. The basic issue is that the dynamics of a Hamiltonian system are highly restricted by conservation laws or volume preservation; violation of these principles by naive schemes (which normally do not take advantage of underlying structure) leads to a gradual corruption of the solution. In particular, the errors may accumulate in such a way that the error in energy is severe. It should be stressed that these issues of instability due to violation of structural features of Hamiltonian systems are not always important in small models or for short time computations (where we can just use very small timesteps); historically, they only became apparent as larger, longer simulations began to be performed, as the appetite for numerical data grew.

The framework of geometric integration builds on an understanding of the properties of Hamiltonian mechanics which are well explained in the book of Arnold [15] or in the monograph of Landau and Lifshitz [212].

2.3.1 Volume Preserving Flows: Liouville's Theorem

Consider a set of points $\mathcal{S}(t)$ in phase space with evolution associated to a differential equation $\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z})$ described by the flow map $\mathscr{F}_t(\mathcal{S}(0)) = \mathcal{S}(t)$. Liouville's theorem [16] states that the volume of such a set is invariant with respect to t if the divergence of $\boldsymbol{f}$ vanishes, i.e.

$$\nabla \cdot \boldsymbol{f} = \sum_{i=1}^{m} \frac{\partial f_i}{\partial z_i} = 0.$$

It is a simple exercise to show that for a Hamiltonian system the divergence vanishes, since

$$\nabla \cdot \boldsymbol{f} = \sum_{i=1}^{\mathrm{N_c}} \frac{\partial^2 H}{\partial q_i \partial p_i} - \sum_{i=1}^{\mathrm{N_c}} \frac{\partial^2 H}{\partial p_i \partial q_i} = 0,$$

by equality of mixed partials. Thus Hamiltonian systems always have volume preserving flows.